WORKED SOLUTION

STEP II 2000, Question 1

Pure • Egyptian Fractions

STEP II 2000, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q1

The key idea

The two examples share a pattern: the first denominator is one more than N, and the second is the product of N and that. For the uniqueness parts, clearing denominators and "completing the rectangle" turns $\frac{1}{a}+\frac{1}{b}=\frac{1}{N}$ into a product of two integers equal to N^2. A prime squared has very few factorisations, so there are very few candidates for a and b, and we can simply list them.

Solution

The general result

In $\frac{1}{2}=\frac{1}{3}+\frac{1}{6}$ and $\frac{1}{3}=\frac{1}{4}+\frac{1}{12}$ the denominators are $N+1$ and $N(N+1)$ (since $6=2\times 3$ and $12=3\times 4$). So we guess

$$\frac{1}{N}=\frac{1}{N+1}+\frac{1}{N(N+1)}.$$

To prove it, put the right-hand side over a common denominator:

$$\frac{1}{N+1}+\frac{1}{N(N+1)}=\frac{N+1}{N(N+1)}=\frac{1}{N}.$$

Both denominators are integers greater than 1, so both terms are unit fractions. They are distinct because $N(N+1)>N+1$ when $N>1$. Hence every unit fraction is a sum of two distinct unit fractions.

Uniqueness when N is prime

Multiplying $\frac{1}{N}=\frac{1}{a}+\frac{1}{b}$ by abN gives $ab=Nb+Na$, that is

$$ab-Na-Nb=0 \quad\Longleftrightarrow\quad ab-Na-Nb+N^2=N^2 \quad\Longleftrightarrow\quad (a-N)(b-N)=N^2.$$

Each step is reversible (as a, b, N are non-zero), so the solutions of $(*)$ are exactly the solutions of $(a-N)(b-N)=N^2$.

Ask yourself Could $a-N$ and $b-N$ both be negative?

No. Since $\frac{1}{b} > 0$, we have $\frac{1}{a} < \frac{1}{N}$, so $a > N$; similarly $b > N$. So $a-N$ and $b-N$ are *positive* integers whose product is N^2. (It's worth noticing: without this, the pair $(-1, -N^2)$ would be another factorisation to rule out.)

Now let $N = p$ be prime. The positive divisors of p^2 are only 1, p and p^2, so the pair $(a-p,\, b-p)$ must be one of

$$(1,\, p^2), \qquad (p,\, p), \qquad (p^2,\, 1).$$

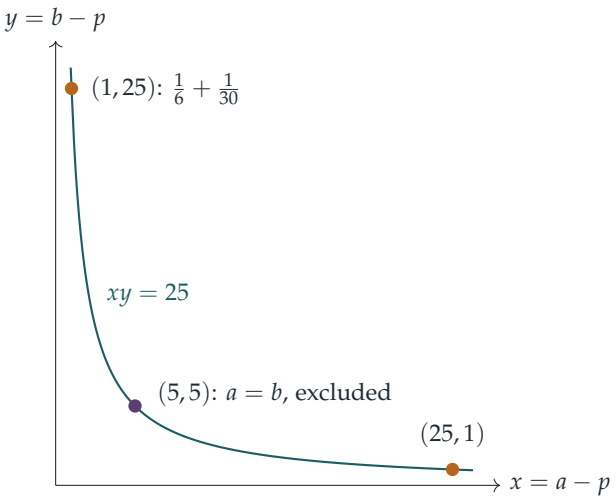
- (p, p) gives $a = b = 2p$, i.e. $\frac{1}{p} = \frac{1}{2p} + \frac{1}{2p}$: the two unit fractions are not distinct, so this is excluded.
- $(1, p^2)$ gives $a = p+1$, $b = p^2 + p = p(p+1)$, which is the representation found above; $(p^2, 1)$ is the same pair in the other order.

So, apart from the order of the two terms, the only way is

$$\frac{1}{p} = \frac{1}{p+1} + \frac{1}{p(p+1)},$$

as required.

The picture shows why: the solutions are the lattice points on the hyperbola $xy = p^2$ in the positive quadrant, where $x = a-p$, $y = b-p$. Here $p = 5$.



Fractions of the form $2/N$

Now suppose $\frac{2}{N} = \frac{1}{a} + \frac{1}{b}$ with $N = p$ an odd prime. Multiplying by abp gives $2ab = pa + pb$. To complete the rectangle we first multiply by 2:

$$4ab - 2pa - 2pb = 0 \quad \Longleftrightarrow \quad 4ab - 2pa - 2pb + p^2 = p^2 \quad \Longleftrightarrow \quad (2a-p)(2b-p) = p^2.$$

Ask yourself Why multiply by 2 before factorising?

Factorising $2ab - pa - pb$ directly would need terms like $(a - \frac{p}{2})$, which aren't integers. Multiplying by 2 keeps everything in whole numbers, so the factors $2a - p$ and $2b - p$ are integers and we can use divisibility.

As before, $\frac{1}{a} < \frac{2}{p}$, so $2a > p$ and $2a - p > 0$; likewise $2b - p > 0$. The positive divisors of p^2 are 1, p, p^2, so $(2a - p, 2b - p)$ is one of

$$(1, p^2), \qquad (p, p), \qquad (p^2, 1).$$

- (p, p) gives $a = b = p$, i.e. $\frac{2}{p} = \frac{1}{p} + \frac{1}{p}$: not distinct, so excluded.
- $(1, p^2)$ gives

  $$a = \frac{p+1}{2}, \qquad b = \frac{p^2 + p}{2} = \frac{p(p+1)}{2}.$$

  Since p is odd, $p + 1$ is even, so both are integers. Since $p \geq 3$, $a \geq 2$, so both are greater than 1, and $b = pa > a$, so they are distinct. The case $(p^2, 1)$ is the same pair in the other order.

Checking directly:

$$\frac{2}{p+1} + \frac{2}{p(p+1)} = \frac{2(p+1)}{p(p+1)} = \frac{2}{p}.$$

So $\frac{2}{p}$ can be written as a sum of two distinct unit fractions, and the only way (up to order) is

$$\frac{2}{p} = \frac{1}{\frac{1}{2}(p+1)} + \frac{1}{\frac{1}{2}p(p+1)},$$

which proves the final result.

Checks

- $p = 3$: $\frac{2}{3} = \frac{1}{2} + \frac{1}{6}$; $p = 5$: $\frac{2}{5} = \frac{1}{3} + \frac{1}{15}$; $p = 7$: $\frac{2}{7} = \frac{1}{4} + \frac{1}{28}$.
- A computer search over all $N < 300$ found exactly one representation of $\frac{1}{N}$ for every prime N and exactly one of $\frac{2}{N}$ for every odd prime N, matching the formulas above.
- Primality really matters: $N = 6$ has four, e.g. $\frac{1}{6} = \frac{1}{7} + \frac{1}{42} = \frac{1}{8} + \frac{1}{24} = \frac{1}{9} + \frac{1}{18} = \frac{1}{10} + \frac{1}{15}$, one for each factorisation $36 = 1 \times 36 = 2 \times 18 = 3 \times 12 = 4 \times 9$ (the pair 6×6 gives $a = b$).
- Why $p > 2$: for $p = 2$, $\frac{2}{2} = 1$ is not of the form $\frac{1}{a} + \frac{1}{b}$ with distinct $a, b > 1$, and indeed $(p + 1)/2$ is not an integer.

Common slips

- Forgetting to show $a - N > 0$ and $b - N > 0$, and so ignoring negative factorisations of N^2.

- Not ruling out the factorisation $N \times N$, which gives equal unit fractions.
- In the $2/N$ case, not checking that $\frac{1}{2}(p+1)$ is an integer (this is exactly where $p \neq 2$ is used).

Beyond STEP

Writing fractions as sums of distinct unit fractions goes back to the ancient Egyptians, and "Egyptian fractions" still hold open problems: the Erdős–Straus conjecture says $\frac{4}{N}$ is always a sum of three unit fractions for $N \geq 2$, and nobody has proved it. The trick of turning $xy - Nx - Ny = 0$ into $(x-N)(y-N) = N^2$ is a standard tool for Diophantine equations, and counting solutions becomes counting divisors. Davenport's *The Higher Arithmetic* is a lovely next step into divisors and primes.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.